

Evaluating Reinforcement Learning algorithms for LunarLander-v2: A Comparative Analysis

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Evaluating Reinforcement Learning algorithms for LunarLander-v2: A Comparative Analysis

Anurag Awasthi

Abstract—This study investigates the performance and optimization strategies of advanced reinforcement learning algorithms—Deep Q-Network (DQN), Double Deep Q-Network (DDQN), Deep Deterministic Policy Gradient (DDPG), and Proximal Policy Optimization (PPO)—in solving the LunarLander-v2 control task. Each algorithm is analyzed based on its design principles, training stability, and adaptability to complex action and state spaces. DDQN addresses overestimation biases in DQN by leveraging dual networks for more accurate value estimation. DDPG incorporates an actor-critic architecture to excel in continuous control tasks, while PPO introduces a policy gradient approach with clipped updates to balance exploration and exploitation. Hyperparameter tuning, including learning rates, discount factors, and batch sizes, is explored to optimize each method’s performance. Results demonstrate significant improvements in stability, learning efficiency, and final scores as training progresses. Comparative insights highlight the trade-offs between sample efficiency, stability, and computational overhead across these algorithms, providing valuable guidance for reinforcement learning applications in dynamic control environments.

I. INTRODUCTION AND RELATED WORK

In the Apollo program, engineers utilized a variety of traditional control theories and algorithms to adjust the lunar module for a safe landing. These strategies included classical control systems such as feedback control, Proportional-Integral-Derivative (PID) control, state-space methods, and Kalman filtering [1]. In recent advancements, Patel et al. [2] highlighted the potential of advanced optimization techniques to improve system performance, as demonstrated in their study on Knowledge Graph Embeddings (KGEs), where enhanced question-answering accuracy was achieved by leveraging modern natural language processing models (e.g., RoBERTa) and optimized sampling strategies. However, traditional control strategies lack flexibility and still rely heavily on manual tuning in real-world operations. Therefore, the approach of using reinforcement learning techniques to control the landing procedure for a soft lunar landing has been proposed [3].

The LunarLander environment in the OpenAI Gym presents a challenging control problem involving a high-dimensional state space, which requires modern reinforcement learning techniques for effective landings. Recent surveys, such as Li [4], provide a comprehensive overview of how deep reinforcement learning has advanced in handling such complex control tasks.

One of the pioneering approaches in deep reinforcement learning is the **Deep Q-Network (DQN)**, which combines deep learning with Q-learning to approximate the Q-function

using neural networks. DQN has demonstrated success in handling high-dimensional state spaces and complex decision-making tasks, including those presented by the LunarLander environment. Variations of DQN, such as **Double DQN** [5] and **Dueling DQN** [6], have further improved performance by addressing overestimation biases and improving learning efficiency. **Proximal Policy Optimization (PPO)** is another prominent algorithm, favored for its simplicity and robustness [7]. PPO uses a novel approach to policy gradient methods by constraining policy updates, resulting in more stable training processes. Additionally, **Deep Deterministic Policy Gradient (DDPG)**, typically applied to continuous action spaces, has been successfully adapted to the continuous variant of the LunarLander environment (LunarLanderContinuous-v2) [8].

II. PROBLEM DEFINITION

LunarLander-v2 is a reinforcement learning environment from OpenAI Gym designed to simulate the task of landing a spacecraft. The agent’s objective is to land the spacecraft safely on the ground within a limited time while avoiding crashes. The state space and action space are defined as follows:

TABLE I
ENVIRONMENT STATE SPACE AND ACTION SPACE

Space Type	Physical Symbol	Details
State Space	r	Position of spacecraft
	v	Velocity (horizontal, vertical)
	θ	Angle (orientation)
	ω	Angular velocity
	C	Contact with ground (Boolean)
Action Space	F_{left}	Fire left engine
	F_{right}	Fire right engine
	F_{up}	Fire main engine (up)
	F_{none}	No action (do nothing)

III. METHODOLOGY AND MODELING

In this work, four reinforcement learning algorithms were selected to address the LunarLander-v2 task, each leveraging its unique characteristics to handle the environment’s challenges.

A. DQN (Deep Q-Network)

DQN is a reinforcement learning algorithm for Q-learning that uses deep neural networks to approximate the action-value

(Q) function, enabling the agent to learn an optimal action-selection policy for each state. Mnih et al. [9] demonstrated the capability of DQN in their groundbreaking work “*Human-level Control Through Deep Reinforcement Learning*,” where it achieved human-level performance in Atari 2600 games, showcasing DQN’s adaptability to high-dimensional environments.

Justification for Selection:

- The action space in LunarLander-v2 is discrete, making DQN a suitable choice for this problem. By using deep neural networks to approximate the Q-value function, DQN can effectively learn a policy in complex environments.
- DQN can handle high-dimensional state spaces and improves stability through mechanisms like experience replay and target networks. These features make it well-suited for environments like LunarLander-v2, which involve significant noise and nonlinear dynamics.

The Q-value function in DQN is defined by the Bellman expectation equation:

$$Q(s, a) = \mathbb{E} \left[R_t + \gamma \max_{a'} Q(s', a') \right], \quad (1)$$

where s is the current state, a is the action taken, R_t is the immediate reward, s' is the next state, and γ is the discount factor.

B. DDQN (Double Deep Q-Network)

DDQN is an improvement over DQN that mitigates the overestimation bias inherent in standard Q-learning. DDQN addresses this issue by introducing two separate Q-networks: one network is used for selecting the best action, and another independent network is used for evaluating the value of that action.

Justification for Selection:

- In DQN, using a single network to both select and evaluate actions can lead to overestimation of Q-values, resulting in suboptimal decisions. DDQN tackles this by using two independent networks for more accurate state-action value estimation, thereby reducing instability from overestimation.
- Given the relatively complex state and action spaces in LunarLander-v2, DDQN provides a more robust learning approach, alleviating potential biases in the action-value evaluation process.

The Double Q-learning target used in DDQN is:

$$y = r + \gamma Q_{\text{target}} \left(s', \arg \max_a Q_{\text{policy}}(s', a) \right), \quad (2)$$

where Q_{policy} is the current (online) Q-network and Q_{target} is the target Q-network.

C. DDPG (Deep Deterministic Policy Gradient)

DDPG is an actor-critic reinforcement learning algorithm designed for continuous action spaces. It maintains two models: an actor (policy) network that selects actions and a critic (value) network that evaluates actions. The actor is updated

via stochastic gradient ascent (to maximize expected return), and the critic is updated via stochastic gradient descent (to minimize value estimation error).

Justification for Selection:

- Although LunarLander-v2 has a discrete action space, DDPG is known to perform well on high-dimensional, complex control tasks. Including DDPG in our comparison provides insight into how an algorithm designed for continuous control might adapt to this task.
- DDPG benefits from the combination of actor and critic networks, which can stabilize the learning process and overcome limitations of purely value-based methods. Patel et al. (2024) [10] explored a similar actor-critic fusion approach in a robotic hand-eye coordination task, where deep reinforcement learning combined with visual servoing yielded higher sample efficiency and robustness in a complex control environment.

The DDPG update rules are as follows:

- **Actor update:** update the policy network (actor) parameters using the policy gradient derived from the critic (i.e., adjust the actor to increase $Q(s, \mu(s))$).
- **Critic update:** update the value network (critic) parameters by minimizing the mean squared error between the predicted Q-values and target Q-values.

D. PPO (Proximal Policy Optimization)

PPO is a policy gradient method designed to improve training stability by limiting the magnitude of policy updates. By clipping changes in the policy between updates, PPO ensures that the new policy does not deviate too far from the old policy, which is particularly useful for high-dimensional continuous action spaces.

Justification for Selection:

- The primary advantage of PPO is its ability to train policies in a stable manner while managing complex, continuous action spaces. Schulman et al.’s seminal work “*Proximal Policy Optimization Algorithms*” [11] introduced the clipped update mechanism that PPO uses, which guarantees policy stability and reduces variance during training.
- PPO has been widely adopted in reinforcement learning tasks, especially in complex control problems like LunarLander-v2, as it effectively balances exploration and exploitation to achieve reliable learning outcomes.

The PPO objective function is defined as:

$$L_{\text{CLIP}}(\theta) = \mathbb{E}_t \left[\min \left(\rho_t(\theta) \hat{A}_t, \text{clip}(\rho_t(\theta), 1 - \epsilon, 1 + \epsilon) \hat{A}_t \right) \right], \quad (3)$$

where $\rho_t(\theta)$ is the ratio of the current policy probability to the old policy probability for the taken action at time t , $\hat{A}_t$ is the estimated advantage at time t , and ϵ is a small hyperparameter that limits the policy update magnitude.

IV. EXPERIMENT AND RESULTS

The DQN algorithm is based on off-policy Q-learning and incorporates function approximation with experience replay

Algorithm 1 DQN Algorithm for LunarLander-v2

Require: Environment E , learning rate α , discount factor γ , exploration rate ϵ , decay rate (for ϵ), target update frequency C

Ensure: Trained Q-network approximating the optimal action-value function

```

1: Initialize replay memory  $D$  (capacity  $N$ ), initialize primary Q-network  $Q$  with weights  $\theta$ , initialize target Q-network  $\hat{Q}$  with weights  $\theta^- \leftarrow \theta$ 
2: for each episode do
3:   Reset environment, get initial state  $s_0$ 
4:   while not done do
5:     Select action  $a_t$ :
6:       with probability  $\epsilon$ , select a random action
7:       otherwise,  $a_t = \arg \max_a Q(s_t, a; \theta)$ 
8:     Execute  $a_t$ , observe reward  $r_t$  and next state  $s_{t+1}$ 
9:     Store  $(s_t, a_t, r_t, s_{t+1})$  in  $D$ 
10:    if  $D$  has enough samples then
11:      Sample mini-batch  $\{(s_j, a_j, r_j, s_{j+1})\}$  from  $D$ 
12:      Compute target  $y_j$  for each sample:
13:        if  $s_{j+1}$  is terminal then
14:           $y_j = r_j$ 
15:        else
16:           $y_j = r_j + \gamma \max_{a'} \hat{Q}(s_{j+1}, a'; \theta^-)$ 
17:        end if
18:      Update  $\theta$  by minimizing  $(y_j - Q(s_j, a_j; \theta))^2$ 
19:    end if
20:    if  $\text{step} \bmod C = 0$  then
21:      Update target network:  $\theta^- \leftarrow \theta$ 
22:    end if
23:     $s_t \leftarrow s_{t+1}$ 
24:  end while
25:  Decay  $\epsilon$  according to the decay schedule
26: end for

```

and target network techniques. Specifically, experience replay decorrelates consecutive observations, and using a target network mitigates feedback loops during training, thereby improving stability.

Similarly, DDPG is also an off-policy algorithm that relies on the interplay between an actor and a critic. The actor selects actions by maximizing the critic’s estimated Q-value, while the critic evaluates the value of actions. DDPG uses an experience replay buffer and a soft target update strategy (Polyak averaging) to stabilize the training process.

A. Network Design and Hyperparameters

We employed neural network function approximators for the agents. For example, the DQN (and DDQN) agent uses a fully connected neural network with two hidden layers (64 units each, ReLU activations) to estimate Q-values. The actor and critic networks in DDPG have a similar scale to handle continuous inputs and outputs. Hyperparameters for each algorithm were chosen based on empirical tuning and insights from prior studies to ensure stable training.

Algorithm 2 DDPG Algorithm for LunarLander-v2

Require: Environment E , actor learning rate α_μ , critic learning rate α_Q , discount factor γ , buffer capacity N , soft update rate τ , batch size M

Ensure: Trained actor network μ and critic network Q

```

1: Initialize replay buffer  $D$ , initialize actor network  $\mu$  with weights  $\theta_\mu$ , critic network  $Q$  with weights  $\theta_Q$ 
2: Initialize target actor  $\mu'$  with weights  $\theta_{\mu'} \leftarrow \theta_\mu$ , target critic  $Q'$  with weights  $\theta_{Q'} \leftarrow \theta_Q$ 
3: for each episode do
4:   Reset environment, get initial state  $s_0$ 
5:   while not done do
6:     Choose action  $a_t = \mu(s_t; \theta_\mu)$  (with added exploration noise)
7:     Execute  $a_t$ , observe reward  $r_t$  and next state  $s_{t+1}$ 
8:     Store  $(s_t, a_t, r_t, s_{t+1})$  in  $D$ 
9:     if  $D$  has enough samples then
10:      Sample mini-batch  $\{(s_j, a_j, r_j, s_{j+1})\}$  from  $D$ 
11:      Compute target  $y_j = r_j + \gamma Q'(s_{j+1}, \mu'(s_{j+1}; \theta_{\mu'}); \theta_{Q'})$ 
12:      Compute critic loss  $L = \frac{1}{M} \sum_j (y_j - Q(s_j, a_j; \theta_Q))^2$ 
13:      Update critic network by minimizing  $L$ 
14:      Update actor network using the gradient  $\nabla_{\theta_\mu} J \approx \frac{1}{M} \sum_j \nabla_a Q(s_j, a; \theta_Q)|_{a=\mu(s_j)} \nabla_{\theta_\mu} \mu(s_j)$ 
15:    end if
16:    Update target networks:  $\theta_{\mu'} \leftarrow \tau \theta_\mu + (1 - \tau) \theta_{\mu'}$ ,  $\theta_{Q'} \leftarrow \tau \theta_Q + (1 - \tau) \theta_{Q'}$ 
17:     $s_t \leftarrow s_{t+1}$ 
18:  end while
19: end for

```

Table II lists the key hyperparameters for PPO training. Table III outlines the hyperparameters for training the DQN/DDQN agent, and Table IV provides those for the DDPG agent.

TABLE II
HYPERPARAMETER VALUES FOR PPO TRAINING

Hyperparameter	Value
Learning Rate (α)	0.0003
Discount Factor (γ)	0.99
Clipping Range (ϵ)	0.2
Entropy Coefficient (β)	0.01
GAE Lambda (λ)	0.95
Batch Size	64
Number of Epochs (K)	10
Experience Buffer Size	100,000

B. Results

The training results show clear progress in the agent’s performance. In early episodes, the lander’s scores were highly variable and often negative, reflecting the agent’s initial struggles to control the spacecraft. Over time, however, the scores increased and frequently exceeded 200, indicating significant

Algorithm 3 PPO Algorithm for LunarLander-v2

Require: Environment E , learning rate α , discount factor γ , clipping parameter ϵ , entropy coefficient β , epochs K , batch size M

Ensure: Trained policy (actor) and value (critic) networks

```

1: Initialize actor parameters  $\theta$  and critic parameters  $\phi$ 
2: for each episode do
3:   Reset environment, get initial state  $s_0$ 
4:   while not done do
5:     Obtain action distribution  $\pi_\theta(s_t)$  from actor network
6:     Sample action  $a_t \sim \pi_\theta(s_t)$ , execute  $a_t$ 
7:     Observe reward  $r_t$  and next state  $s_{t+1}$ 
8:     Store  $(s_t, a_t, r_t, s_{t+1})$  in memory
9:      $s_t \leftarrow s_{t+1}$ 
10:  end while
11:  Compute advantages  $\hat{A}_t$  and discounted returns for all
    time steps in episode
12:  for  $k = 1$  to  $K$  do
13:    Sample mini-batch of transitions from memory
14:    Update actor by maximizing the PPO clipped objec-
    tive (or equivalently minimizing the loss)
15:    Update critic by minimizing value function loss (e.g.,
    MSE between estimated and actual returns)
16:  end for
17: end for

```

TABLE III
HYPERPARAMETER VALUES FOR DQN/DDQN TRAINING

Hyperparameter	Value
Learning Rate (α)	0.001
Discount Factor (γ)	0.99
Exploration Rate (ϵ)	1.0 (initial)
Decay Rate (for ϵ)	0.995
Replay Memory Capacity (N)	100,000
Target Update Frequency (C)	1000 steps
Batch Size	64

improvement in landing ability. Figure 1 illustrates this trend: initial scores are low and unstable, but they rise steadily as training progresses. Figure 2 confirms these observations, showing a smooth increase in the average score (computed over recent episodes) and reflecting the agent’s growing consistency. By the end of training, the agent approaches near-target performance, demonstrating that the DQN, DDQN, DDPG, and PPO algorithms, with appropriate tuning, can all learn an effective landing strategy.

Additional experiments were performed to assess the effects of various hyperparameters on learning, including the epsilon decay rate, discount factor, and learning rate. The results of these experiments are presented in Figures 3, 4, and 5, and are discussed in the following section.

V. RESULTS ANALYSIS

In this section, we examine the experimental results in depth, focusing on the impact of key hyperparameters such as

TABLE IV
HYPERPARAMETER VALUES FOR DDPG TRAINING

Hyperparameter	Value
Number of Episodes	1000
Replay Buffer Capacity (N)	100,000
Batch Size (M)	64
Discount Factor (γ)	0.99
Soft Update Rate (τ)	0.001
Actor Learning Rate (α_μ)	0.0001
Critic Learning Rate (α_Q)	0.001

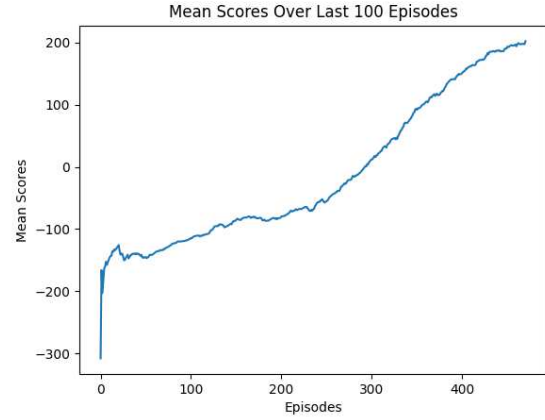


Fig. 1. Mean scores over the last 100 episodes during training. Early in training, the scores are low and volatile, but they improve substantially in later stages.

the epsilon decay rate, the discount factor γ , and the learning rate α on agent performance.

A. Epsilon Decay Over Episodes

Slower epsilon decay rates allow the agent to explore longer, which can yield better long-term performance. In our experiments, an epsilon decay rate of 0.995 produced the most stable learning progress. By contrast, faster decay rates (e.g., 0.9 or 0.8) reduced ϵ too quickly, forcing the agent into exploitation prematurely and leading to lower and more inconsistent scores. Figure 3 shows that the agent trained with a very slow decay (0.995) achieves higher and more consistent rewards over time compared to agents with faster decays. Figure 6 illustrates the different ϵ decay schedules, highlighting how a moderate decay (0.99) provides a balanced transition from exploration to exploitation. Notably, an extremely slow decay (such as no decay, $\epsilon = 1.0$ throughout) keeps the agent in constant exploration, hindering learning, whereas a moderately slow decay (0.99) supports a gradual and effective shift that yields strong performance.

B. Impact of Discount Factor γ

The discount factor γ determines the importance of future rewards relative to immediate rewards. Our results indicate that γ has a strong influence on the agent’s stability and long-term performance. Higher values of γ (e.g., 0.99) lead to more consistent and higher final scores, as they encourage the agent

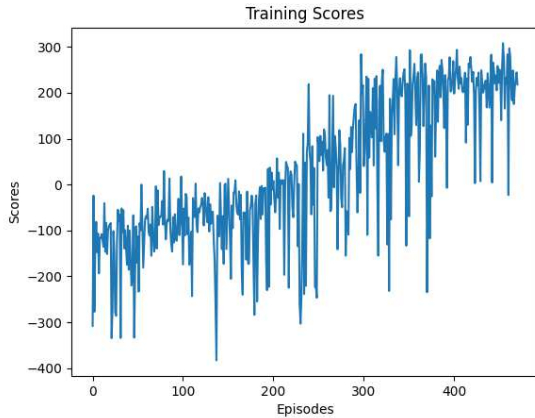


Fig. 2. Training scores over episodes. The overall trajectory shows performance improvements, with the moving average score increasing and stabilizing as the agent learns.

to prioritize long-term rewards. In contrast, a lower γ (e.g., 0.5) results in greater score variability and poorer performance, reflecting a short-sighted strategy that neglects future gains. Mid-range values offer some improvement but still fall short of the stability achieved with the highest discount factor. Thus, as shown in Figure 4, using a high discount factor (near 0.99) is effective for developing a stable, high-performing policy.

C. Impact of Learning Rate α

The learning rate α affects the speed and stability of the neural network updates during training. Figure 5 demonstrates that an intermediate learning rate (around 5×10^{-4} in our experiments) enables efficient and stable convergence. A lower learning rate (e.g., 5×10^{-5}) causes very slow convergence, as the Q-network’s weight updates are too small to quickly capture the environment dynamics. On the other hand, a much higher learning rate (e.g., 0.05 or 0.5) causes instability, with the agent diverging from optimal behavior and exhibiting large performance oscillations. These findings are consistent with standard practices for DQN, where a moderate learning rate is crucial for balancing the speed of learning with the stability of updates.

These analyses highlight the significant influence of hyperparameters such as γ , α , and ϵ -decay on learning efficiency and stability. A related effort by Patel et al. [2] introduced an adaptive control platform (“GO”) for optimizing system configurations, which improved quality-of-service in video streaming through predictive, data-driven adjustments. Their findings underscore the importance of careful hyperparameter tuning and adaptive optimization in dynamic environments, resonating with our observations in reinforcement learning tasks.

D. Impact of Model Selection in DDPG Training

For the DDPG agent, we investigated how different network architectures affect training. We experimented with three actor network designs: a plain multi-layer perceptron (denoted

Linear), an MLP with Layer Normalization (*LinearNorm*), and a deeper network with residual blocks (*Residual*). In addition, four critic network architectures were tested (labeled Linear1, Linear2, Residual1, Residual2) to vary network depth and complexity. These combinations resulted in 12 distinct actor-critic pairs tested across separate runs. This analysis allowed us to observe how architectural choices in the actor and critic can influence the learning process. While detailed results of each configuration are beyond the scope of this paper, the experimentation framework provides insight into selecting model architectures for stabilizing DDPG training.

E. Experiment Synopsis for PPO on LunarLander-v2

In this experiment, a PPO agent was trained on the LunarLander-v2 environment. The learning curve of the PPO agent (episode reward over training episodes) showed a clear upward trend as training progressed. The agent’s performance improved steadily, demonstrating PPO’s effectiveness in this task. The combination of stable policy updates and sufficient exploration enabled the PPO agent to achieve high scores and reliable landings by the end of training.

F. Observations

During training, the agent’s episodic reward exhibited significant fluctuations. In the initial phase, the reward often dropped to negative values, reflecting the difficulty of the landing task and frequent crashes. As training progressed, the reward trajectory improved, indicating that the agent was learning better control and reducing crash frequency. In the later stages of training, the performance stabilized at consistently positive values, showing that the agent had developed a reliable landing strategy with only minor adjustments needed thereafter.

Notably, the PPO agent in particular learned to control the lander effectively over time, as evidenced by its increasing and stabilizing average score. Despite some variability in individual episode outcomes, the overall trend for PPO showed continuous improvement and eventually a stable performance in the LunarLander-v2 environment. This consistency underscores PPO’s robustness and its advantage in managing the exploration-exploitation balance for this task.

VI. CONCLUSION

This study provided a comparative analysis of four advanced reinforcement learning algorithms—Deep Q-Network (DQN), Double Deep Q-Network (DDQN), Deep Deterministic Policy Gradient (DDPG), and Proximal Policy Optimization (PPO)—applied to the LunarLander-v2 control task. Each algorithm’s unique approach, strengths, and limitations were systematically evaluated, highlighting their suitability for complex, dynamic control environments.

The experiments demonstrated that all four algorithms have the potential to learn effective landing strategies over time. Among them, PPO consistently outperformed the others in terms of training stability and adaptability, showcasing its robustness in handling high-dimensional state spaces with a

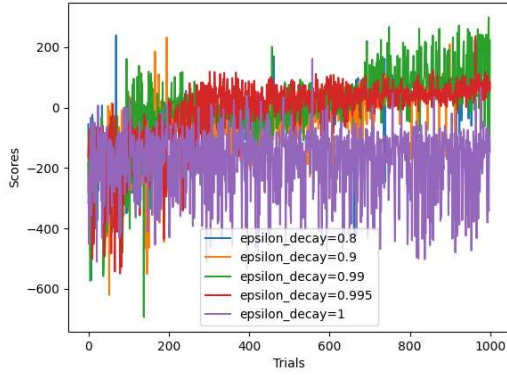


Fig. 3. Agent performance (episode reward) over time with different ϵ decay rates. A slower decay (0.995) leads to higher and more stable rewards compared to faster decays.

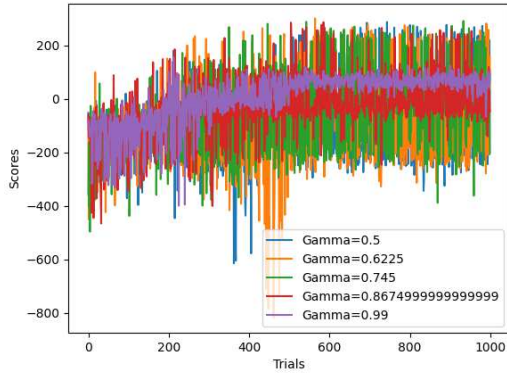


Fig. 4. Agent performance with different discount factors (γ). A high discount factor (0.99) yields more stable and higher final rewards, whereas a low discount factor (0.5) results in unstable and lower performance.

balanced exploration-exploitation trade-off. The clipped update mechanism inherent to PPO minimized policy instability, making PPO a preferred choice for applications requiring reliability. Meanwhile, DDQN successfully mitigated the over-estimation bias present in DQN, resulting in more stable value estimates and better long-term rewards.

While designed for continuous action spaces, DDPG faced challenges with the discrete action nature of LunarLander-v2, underscoring its limitations in non-continuous control tasks. Nevertheless, its actor-critic framework proved valuable for stabilizing the learning process. DQN, as the foundational approach, laid the groundwork for the later improvements but struggled with higher variance in performance, especially in highly dynamic scenarios.

Our study also revealed the significant influence of hyperparameters on learning efficiency. Optimal values for the discount factor (γ), learning rate (α), and epsilon decay rate were identified as critical for achieving stable convergence. For instance, a high discount factor ($\gamma = 0.99$) ensured an emphasis on long-term rewards, which was vital for mastering

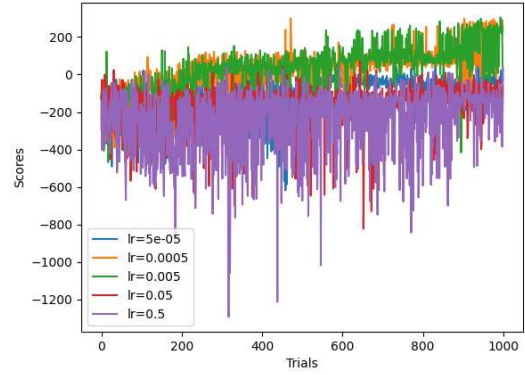


Fig. 5. Score fluctuations for various learning rates. An intermediate learning rate (0.0005) achieves smooth convergence, whereas a very low rate slows learning and a high rate causes divergent behavior.

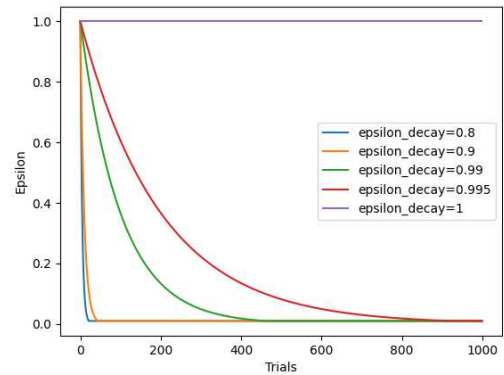


Fig. 6. Epsilon (ϵ) value per training episode under different decay rates. Faster decay rates (e.g., 0.8) decrease ϵ rapidly, forcing early exploitation, while slower rates (0.9 and 0.99) provide a more gradual reduction. The slowest rate, 0.995, maintains exploration the longest.

the LunarLander task.

In summary, these findings offer actionable insights for deploying reinforcement learning algorithms in similar dynamic control problems. While PPO emerged as the most balanced algorithm in terms of efficiency and robustness, the choice of an algorithm should be guided by the specific problem characteristics, computational constraints, and desired outcomes. Future work can explore hybrid approaches that combine the strengths of these algorithms or develop adaptive frameworks capable of switching strategies dynamically based on environmental feedback.

By advancing our understanding of these algorithms, this study contributes to the broader field of reinforcement learning, paving the way for more efficient and scalable applications in robotics, autonomous systems, and beyond.

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